Practical No: 7-C

Aim: Develop an Android application using Adapters to populate Spinner or ListView dynamically.

Spinner

A Spinner is a dropdown list that allows users to select one item from multiple options.

Adapter

An Adapter acts as a bridge between the data source and the UI component.

ArrayAdapter

An ArrayAdapter is used to connect an array of data to a Spinner or ListView.

activitymain.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity"
    android:orientation="vertical"
    android:gravity="center">

    <Spinner
        android:id="@+id/spinnerSubjects"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

</LinearLayout>
```

MainActivity.java

```
package com.example.pract1;

import android.annotation.SuppressLint;
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.MotionEvent;
import android.view.View;
import android.widget.ArrayAdapter;
import android.widget.Button;
import android.widget.ImageView;
import android.widget.Spinner;
import android.widget.TextView;
```

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```
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

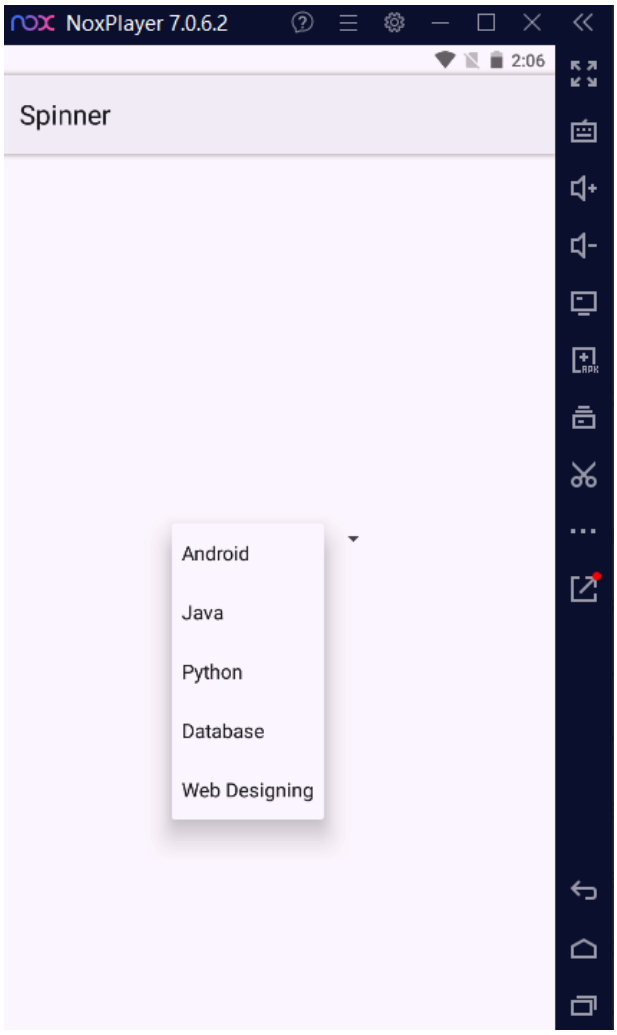
public class MainActivity extends AppCompatActivity {
  Spinner spinnerSubjects;
  String[] subjects = {
        "Android",
        "Java",
        "Python",
        "Database",
        "Web Designing"
  };
  @SuppressLint("MissingInflatedId")
  @Override
  protected void onCreate(Bundle savedInstanceState) {
     super.onCreate(savedInstanceState);
     EdgeToEdge.enable(this);
     setContentView(R.layout.activity_main);
spinnerSubjects =
           findViewById(R.id.spinnerSubjects);
     ArrayAdapter<String> adapter =
           new ArrayAdapter<>(
                this,
                android.R.layout.simple_spinner_item,
                subjects);
     adapter.setDropDownViewResource(
           android.R.layout.simple_spinner_dropdown_item);
     spinnerSubjects.setAdapter(adapter);
        spinnerSubjects.setOnItemSelectedListener(
                  new AdapterView.OnItemSelectedListener() {
                      @Override
                      public void onItemSelected(AdapterView<?> parent,
View view, int position, long id) {
                   Toast.makeText(
                                 MainActivity.this,
                                 "Selected: " + subjects[position],
                                 Toast.LENGTH_SHORT
                         ).show();
                      }
```

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```
        @Override
        public void onNothingSelected(AdapterView<?> parent) {

                        }
    });
}
}
```

Output:



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